

An Internship Report
Of
Google AI-ML Virtual Internship

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech (Computer Science and Engineering)

By

Name of Student: SHREYANSH SRIVASTAVA

Roll Number: Not provided

Batch: Not provided

CANDIDATE'S DECLARATION

I, SHREYANSH SRIVASTAVA, do hereby declare that the internship report titled “Google AI-ML Virtual Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-

Student Name: SHREYANSH SRIVASTAVA

Roll No.: Not provided

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the AICTE – EduSkills Virtual Internship Program team, EduSkills Academy, and Google for Developers for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of Google AI-ML. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature:-

Student Name: SHREYANSH SRIVASTAVA

Roll No.: Not provided

TABLE OF CONTENTS

1. Candidate's Declaration
2. Acknowledgement
3. Internship Completion Certificate
4. Project Description
 - 4.1 Introduction
 - 4.2 Organization Profile
 - 4.3 Problem Statement
 - 4.4 Project Objectives
 - 4.5 Scope of the Project
 - 4.6 Technologies and Tools Used
 - 4.7 System Architecture
 - 4.8 Methodology
 - 4.9 Expected Outcomes
 - 4.10 Certificates of Completion and Communication Proof
5. Bibliography/References

3. INTERNSHIP COMPLETION CERTIFICATE

**शिक्षा मंत्रालय
MINISTRY OF
EDUCATION**
सत्यमेव जयते

**AICTE**
All India Council for Technical Education
अखिल भारतीय तकनीकी शिक्षा परिषद्

**NATIONAL
INTERNSHIP
PORTAL**

**EduSkills®**
Nation Building Through Skills



Certificate of Virtual Internship

Shreyansh Srivastava
IILM University, Greater Noida
has successfully completed the 10-weeks
AI-ML Virtual Internship
During December 2025 to May 2026
May this Internship learning propel you toward a bright and successful career.

Supported By : **India Edu Program**




Karthik Padmanabhan
Developer Ecosystem Lead
MENA & India, Google



Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education



Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills



Certificate ID : a7f3e6c9d4b2f1e8a6c3d5b7
Student ID : shreyansh.srivastava.cs28@iilm.edu



GRADE: O (Outstanding: 90-100) | E (Excellent: 80-89) | A (Very Good: 70-79) | B (Good: 60-69) | C (Fair: 50-59) | D (Average: 40-49) | P (Pass): 30-39 | F (Fail): Below 30

Certificate Details

Field	Details
Name	Shreyansh Srivastava
Institute	IILM University, Greater Noida
Internship	AI-ML Virtual Internship
Duration	10 Weeks

Field	Details
Period	December 2025 – May 2026
Grade	O (Outstanding)
Certificate ID	a7f3e6c9d4b2f1e8a6c3d5b7
Student ID	shreyansh.srivastava.cs28@iilm.edu
Supported By	India Edu Program / Google for Developers

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the 10-weeks Virtual Internship undertaken under the AICTE – EduSkills Virtual Internship Program, in the domain of AI-ML. The certificate records the internship period as December 2025 to May 2026 and identifies IILM University, Greater Noida as the student's institute. The internship was supported by the India Edu Program / Google for Developers.

The learning focus covered practical foundations of Machine Learning and Deep Learning, with emphasis on TensorFlow-based computer-vision workflows including image classification, object detection, on-device machine learning, and product image search.

4.2 Organization Profile

EduSkills Academy operates the AICTE – EduSkills Virtual Internship Program under the patronage of the AICTE National Internship Portal. The Google AI-ML learning track is supported by Google for Developers and is designed around structured learning pathways, hands-on activities, quizzes, and skill-based learning.

4.3 Problem Statement

Although engineering students may learn theoretical foundations of machine learning and neural networks during academic coursework, practical exposure to building and applying computer-vision models using an industry-standard framework such as TensorFlow can be limited. The internship addressed this learning gap through a structured workflow covering neural networks, image classification, object detection, TensorFlow Lite, and application integration.

4.4 Project Objectives

- To understand the first principles of machine learning and build convolutional neural networks (CNNs) using TensorFlow for image recognition and classification.
- To learn the basics of object detection and integrate pretrained object detectors into mobile applications.
- To train custom object-detection models using TensorFlow Lite and the TFLite Model Maker.
- To build a product image search feature using on-device object detection.
- To detect objects in static images and live camera feeds for visual product search.
- To understand backend integration for a product image search mobile application.
- To build custom image-classification models and improve classification skills.
- To create and integrate a custom image-classifier model into an application.
- To complete the internship requirements and achieve the final credential validation.

4.5 Scope of the Project

The scope covers a TensorFlow-based computer-vision workflow beginning with neural-network fundamentals and CNNs, followed by on-device object detection using pretrained and custom models. It also covers product image search using static images and live camera feeds, custom image classification, and integration of trained models into an application. The work described here represents structured virtual learning and lab-based activities; the supplied certificate does not document deployment on a live production system.

4.6 Technologies and Tools Used

- Python
- TensorFlow
- Keras
- TensorFlow Lite
- TensorFlow Lite Model Maker
- Convolutional Neural Networks (CNNs)
- Object Detection
- Image Classification
- Mobile Application Integration
- Google Developer Program learning resources

4.7 System Architecture

- **Model Layer:** A convolutional neural network built and trained using TensorFlow forms the core intelligence layer for image recognition and classification.
- **Conversion Layer:** Trained models can be converted into lightweight on-device formats using TensorFlow Lite and the TFLite Model Maker.
- **Detection Layer:** Pretrained and custom-trained object detectors identify and localize objects in images and camera feeds.
- **Search Layer:** Product image search combines object detection with application/backend functionality for visual search.
- **Classification Layer:** A custom image classifier categorizes images or detected objects.
- **Application Layer:** Trained and converted models are integrated into an application to provide detection, search, and classification capabilities.
- **Assessment Layer:** Learning activities, quizzes, badges, and final credential validation are used to assess progress.

4.8 Methodology

The internship followed a structured learning methodology combining conceptual study, hands-on activities and assessments. The source report used an eight-module curriculum; the certificate supplied for Shreyansh records the overall internship duration as 10 weeks. The module-wise learning sequence is reproduced below because the supplied report establishes these as the Google AI-ML learning modules.

Module	Topic	Description
1	Program Neural Networks with TensorFlow	Foundational concepts of machine learning and building CNNs for image recognition and classification.
2	Get Started with Object Detection	Basics of object detection and integrating pretrained object detectors into applications.
3	Go Further with Object Detection	Training custom object-detection models using TensorFlow Lite and the TFLite Model Maker.
4	Get Started with Product Image Search	Building a product image search feature using on-device object detection.
5	Get Started with Product Image Search	Detecting objects in static images and live camera feeds for visual product search.
6	Go Further with Product Image Search	Backend integration for a product image search mobile application.
7	Go Further with Image Classification	Building custom image-classification models and improving classification skills.

Module	Topic	Description
8	Go Further with Image Classification	Creating and integrating a custom image-classifier model into an application.
Final	Final Credential Validation	Final credential validation / internship assessment.

Note: The supplied Shreyansh certificate records a 10-weeks internship, while the source report used for the structure describes the Google AI-ML curriculum as eight learning modules. No separate Shreyansh weekly assessment record was supplied with the files.

4.9 Expected Outcomes

- Ability to explain and apply machine-learning fundamentals to build convolutional neural networks using TensorFlow.
- Practical understanding of object detection using pretrained and custom TensorFlow Lite models.
- Hands-on understanding of product image search combining on-device detection with application/backend integration.
- Capability to build, evaluate, and improve custom image-classification models.
- Ability to integrate trained and converted TensorFlow Lite models into an application.
- Completion of the internship and achievement of an “Outstanding” grade as stated on the supplied certificate.

4.10 Certificates of Completion and Communication Proof

The Internship Completion Certificate supplied for Shreyansh Srivastava confirms successful completion of the AI-ML Virtual Internship. The certificate states a 10-weeks duration during December 2025 to May 2026 and an overall grade of O (Outstanding). The certificate also contains a QR code for verification. No separate offer letter, completion email, or Google Developer Program badge screenshot for Shreyansh was supplied with the current files, so those items have not been reproduced or fabricated in this report.

5. BIBLIOGRAPHY/REFERENCES

1. EduSkills Academy, “AICTE – EduSkills Virtual Internship Program – AI-ML,” Internship Completion Certificate, 2026.
2. Google Developers, Machine Learning documentation, <https://developers.google.com/machine-learning>
3. Google Developer Program, <https://developers.google.com/profile>
4. TensorFlow Documentation, <https://www.tensorflow.org/>
5. TensorFlow Lite Documentation, <https://www.tensorflow.org/lite>
6. TensorFlow Lite Model Maker Documentation, https://www.tensorflow.org/lite/models/modify/model_maker
7. AICTE National Internship Portal, <https://internship.aicte-india.org/>
8. EduSkills Foundation, <https://eduskillsfoundation.org/login>